

deliver a tailored and multi-dimensional solution for students.

Little Dragon: This platform employs behavioural and emotional aspects of learning. An emotional AI detects states like boredom and frustration, and then adjusts content to optimise the experience of learning.

Brainly: This AI-based solution offers personalised materials. It is basically a social learning community to help students. They can discuss issues related to their homework or gain new knowledge from other students. This platform utilises ML to provide a better user experience and assists in identifying spam and inappropriate content.

Carnegie Learning: This is an online learning platform for middle and high school students. This system tends to provide more customised education materials, making the learning process more comfortable. It offers real-time education, analyses users' keystrokes, and allows the tutor to see students' progress.

Third Space Learning: Here, AI is used to monitor lessons each week, pair students with their ideal lesson, and pay attention to tutors and the way they teach lessons. The system can recommend ways to improve teaching techniques. For example, if the teacher speaks too fast or too slow, the system sends a notification.

Advantages for students

Learn at any time: Many AI-based apps for learning are available online. These apps enable students to learn lessons on the go or study in their free time. Moreover, students can get feedback from tutors in real-time mode.

Flexible options: AI-based solutions are available depending on students' level of knowledge, topics, and choices. For example, the system lets students take an initial test before using the app. The app analyses the test result and provides suitable tasks and courses.

Virtual mentors: AI-based platforms offer virtual mentors to track students' progress. It's good to get instant feedback from the virtual tutor before human teachers can evaluate the actual progress.

Advantages for teachers

Quick assessment: Different AI-based online courses allow teachers to quickly assess a student's knowledge. For example, if most of the students gave wrong answers to a particular question in an online test, the teacher can pay more attention to that particular topic.

Better engagement: Teachers often find it difficult to create a productive engagement within the class. Many modern technologies such as VR classrooms help students to get involved in the education process, making them more interactive. With VR technology, students become more engaged, talk about their experiences, and feel more motivated to learn. It also helps teachers to keep their students engaged in the courses.

Personalisation: Various AI-enabled algorithms can analyse students' interests and provide more personalised recommendations and training programmes.

AI in Indian education space

The government has set up a task force to boost the AI sector, including AI in education and research industries. The National Association of Software and Services Companies (Nasscom), in collaboration with government's NITI Aayog, has launched an AI-based module for students of Indian schools.

India's Central Board of Secondary Education (CBSE) has recently introduced two handbooks for teachers to help integrate AI in schools. The first is about AI curriculum and experimental methodologies covering both social and technological skills. The second is about AI integration across subjects to enhance the

multidisciplinary approach in the teaching-learning process. This handbook discusses how AI-based tools can enhance learning within and outside classrooms.

Many Indian institutes are utilising AI to enhance their learning processes and provide quality education to students. Apart from homegrown video-conferencing apps such as JioMeet and Say Namaste, there are many AI-based learning platforms launched by Indian startups. Some of the use cases launched by Indian companies are:

Thinkster Maths: It is an AI-based math learning platform that combines a world-class curriculum with personalisation. The system tracks and visualises how a student is thinking while he or she works on a problem. This helps teachers to quickly spot problem areas of the student. It was founded by alumni from IIT and BITS.

BYJU'S: This popular online class in India provides educational content mainly to school students from classes one to twelve. It employs AI, ML, computer vision, cloud-based analytics, as well as augmented and virtual reality. BYJU'S features include personalised and interactive learning and in-depth analysis. It was founded in 2011 by Byju Raveendran.

VideoKen: This AI-based platform makes informational videos richer, consumable, and engaging. It works on cutting edge technologies involving Big Data analytics, AI, and ML. It offers an online video-based social learning platform. Users can search and curate educational video content based on the required topic, quickly search the part of the video they require based on phrases, store it as a private note, and share it with others. It was founded in January 2016 by Manish Gupta.

Utter: It is a chatbot-based application available on an Android device for learning English.

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